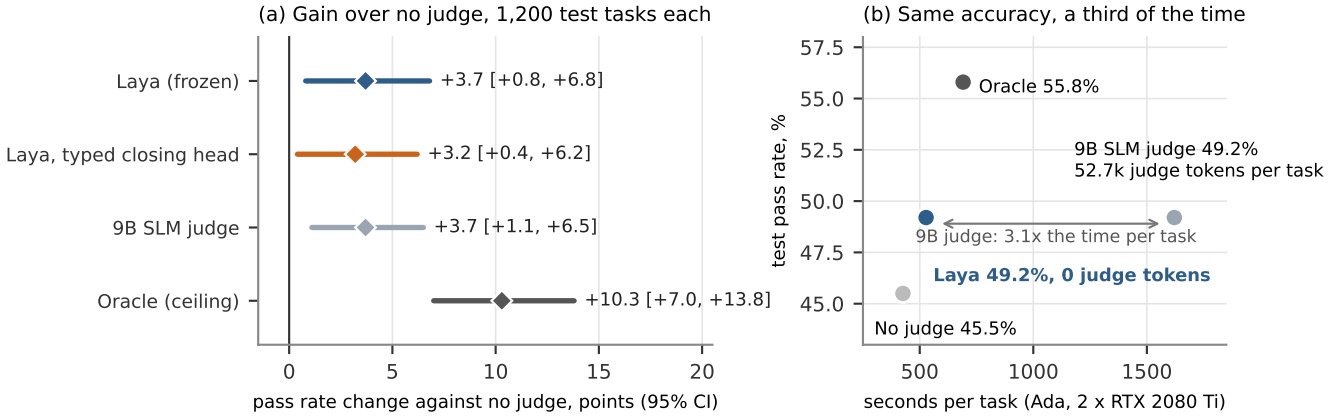
(a) Gain over no judge, 1,200 test tasks each
Laya (frozen)
+3.7 [+0.8, +6.8]
Laya, typed closing head
+3.2 [+0.4, +6.2]
9B SLM judge
+3.7 [+1.1, +6.5]
Oracle (ceiling)
+10.3 [+7.0, +13.8]
0
5
10
15
20
pass rate change against no judge, points (95% CI)
(b) Same accuracy, a third of the time
test pass rate, %
57.5
55.0
52.5
50.0
47.5
45.0
Oracle 55.8%
9B SLM judge 49.2%
52.7k judge tokens per task
9B judge: 3.1x the time per task
Laya 49.2%, 0 judge tokens
No judge 45.5%
500
1000
1500
seconds per task (Ada, 2 x RTX 2080 Ti)